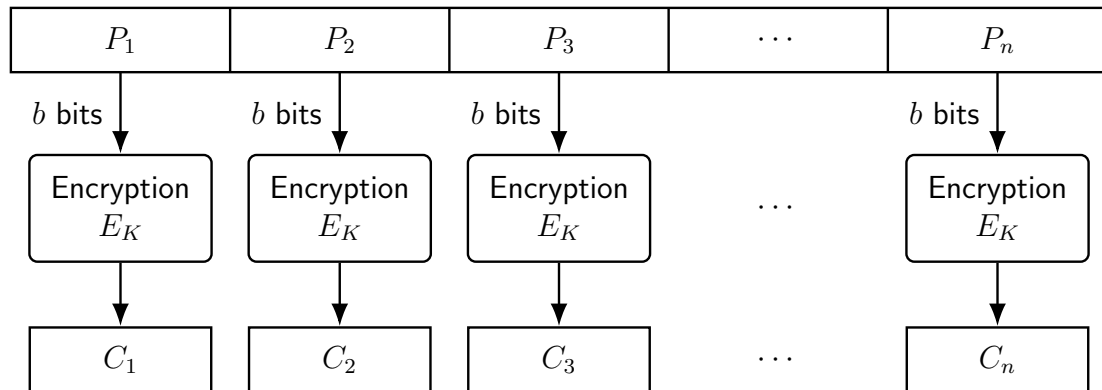


Block Cipher: Basic Concept

Plaintext stream partitioned into b -bit blocks



Every E_K uses the same secret key K

Fixed-size blocks $P_i, C_i \in \{0, 1\}^b$

$$C_i = E_K(P_i)$$

Single-block transformation shown. A mode of operation specifies how the block cipher is used to encrypt a complete message.